

Savita Pareek

Postdoctoral Associate

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EDUCATION	Ph.D., Statistics, Indian Institute of Technology Bombay, India 2018 - 2023 <i>Dissertation:</i> On Some Problems in Mixed-effect Models, CPI: 9.0/10 <i>Advisor:</i> Prof. Siuli Mukhopadhyay M.Sc., Mathematical Sciences - Statistics 2014 - 2016 Banasthali University, India, Percentage Marks: 86.72/100 B.Sc., Mathematics, Physics, and Computer Science 2011 - 2014 Banasthali University, India, Percentage Marks: 84.06/100
PROFESSIONAL EXPERIENCE	Postdoctoral Associate, Massachusetts Institute of Technology, USA Jan 2026 – Present MIT Sloan School of Management and Schwarzman College of Computing (SERC) <i>Mentors:</i> Prof. Rahul Mazumder, Prof. Suzanne Flynn • Developing Interpretable ML methods for linguistic data processing to diagnose Alzheimer's. • Cohort leader for the Social and Ethical Responsibilities of Computing (SERC) program. Postdoctoral Research Fellow, Auburn University, USA Aug 2024 – Dec 2025 Department of Mathematics and Statistics; <i>Mentor:</i> Prof. Roberto Molinari • Developed differentially private equivalence tests for two-sample means and proportions. • Established simulation-calibrated validity under privacy constraints and implemented them in the cTOST R package. • Multiple-Principal Investigator on an NIH R21 proposal leading grant development. • Proposed a Rashomon-based inference for mental health heterogeneity, including ADHD. Research Officer & Manager, Reserve Bank of India (RBI), Mumbai Feb 2021 – Aug 2024 Department of Statistics and Information Management • Designed an International Financial Services Centres (IFSC) foreign exchange information-flow system, issuing an official circular standardizing FX transactions and positions. • Led compilation and validation of millions of monthly transaction-level records for India's official Balance of Payments statistics. Ph.D. Research Projects, Indian Institute of Technology Bombay, India Jan 2018 - Oct 2023 • Mixed-effects models for crossover trials with missing and skewed responses, using Monte Carlo EM, applied to gene expression data from a 3×3 study. • Model-based clustering of gene expression time series, capturing temporal heterogeneity via mixed-effects models and Dirichlet process priors, applied to <i>Streptomyces coelicolor</i>.
TEACHING & MENTORSHIP	Instructor Auburn University, USA • Fundamentals of Statistical Inference I (STAT 5510/6610, 11 students) Fall 2025 New graduate-level course; prepared original lectures, assignments, and assessments. Teaching evaluation (out of 6.0): mean 5.8, median 6.0. • Introduction to Data Science (STAT 4000, 46 students) Spring 2025 Second departmental offering; refined lectures, projects, and assessments. Teaching evaluation (out of 6.0): mean 3.2. • Probability & Statistics II (STAT 3610, 21 students) Fall 2024 Teaching evaluation (out of 6.0): median 4.0. Teaching Assistant Indian Institute of Technology Bombay, India • Introduction to Probability (SI 427) Fall 2020 Led weekly tutorials; graded assignments and exams. • Regression Analysis (SI 422) Spring 2019, 2020 Conducted R programming labs; assisted student projects; graded exams. Mentorship • Undergraduate student, Honors Contract project Spring 2025 • Graduate student, Rashomon sets in multi-model inference 2024-2025

- Research intern, India's current account sustainability Spring 2021
- Summer interns, Power BI dashboard, and the services trade Summer 2022, 2023
- Tutored Science & Math to students from underserved communities 2018-19

JOURNAL PUBLICATIONS

Pareek, S. and Ghosh, S. (2026). *Semiparametric Dynamic Copula Models for Portfolio Optimization*. Journal of Data Science, Minor revision. [copula]

Pareek, S., Insolia, L., Molinari, R., and Guerrier, S. (2026). *Equivalence Testing Under Privacy Constraints*. Under review (Statistics in Medicine)

Pareek, S., Das, K., and Mukhopadhyay, S. (2025). *Likelihood-based inference for skewed responses in a crossover trial setup*. Communications in Statistics - Simulation and Computation. [cssc]

Eady, S., Ulloa, I. N., **Pareek, S.**, and Zabala, M. (2025). *The Effects of Psychosocial Stress on Surface Electromyography Signals of the Lower Leg During Isometric Contractions*. Submitted

Pareek, S., Das, K., and Mukhopadhyay, S. (2023). *Likelihood-based missing data analysis in crossover trials*. Brazilian Journal of Probability and Statistics. [bjps]

Pareek, S., Khetan, M., Mukhopadhyay, S., and Das, K. (2021). *Clustering gene expression time series data embedded in a non-parametric setup*. JSR (Special Issue). [jsr]

POLICY & BOOK CHAPTERS

Misra, A., **Pareek, S.**, and Saggar, S. (2024). *Capturing e-commerce transactions from the travel group of BPM6: An Indian experience*. In External statistics in a fragmented and uncertain world (Vol. 62). [BIS]

Biswas, D., **Pareek, S.**, and Saggar, S. (2022). *Freight costs of India's trade*. RBI Bulletin. [RBI]

GRANTS & FUNDING

NIH R21 Proposal, MPI: **Pareek, S.**, and Molinari, R.; Co-PI: Deshpande, G. *Personalizing Mental Disorder Diagnosis with Rashomon Inference*. Submitted, Oct 2025.

SELECTED WORK IN PROGRESS

Pareek, S., Voirol, L., and Molinari, R. *Towards Scalable Gaussian Process Regression with a Moment based Estimation Procedure*.

Loureiro-Ferreira, T., Voirol, L., **Pareek, S.**, Montillet, J.-P., and Guerrier, S. *Hypothesis Testing for Tectonic Velocities from Spatially Correlated GNSS Time Series*.

HONORS & AWARDS

Research Awards

- Early Career Scholarship, Symposium on Data Science and Statistics (SDSS) 2025
- National Science Foundation (NSF) Travel Award, IISA Conference 2024
- Young Scientist Award, Society of Statistics and Computer Applications (SSCA), India 2024
- CWS Prof. Suddhendu Biswas Student Travel Award 2023
- International Indian Statistical Association (IISA) Student Travel Award 2022
- Gold Medal for Best Student in Science, Banasthali University, India 2016

Professional Activity Awards

- Certificate of Excellence, RBI Induction Program, India 2023
- Institute Organizational Special Mention, for IITB Hostel General Secretary leadership 2021
- Institute Sports Color, IIT Bombay, for excellence in competitive sports 2021
- Best Office Bearer, IITB Hostel 11, for exemplary service to the hostel community 2020

Competitive Exams

- 3rd in RBI Officer Exam (Statistics): among 200k 2020
- 6th in GATE (Mathematics): among 6k 2017
- 24th in CSIR-NET (Mathematical Sciences): among 40k 2017

PRESENTATIONS

Invited Talks

- CFE-CMStatistics Conference, London (virtual) Dec 2025
- Seminar Speaker, Clemson University (virtual) Sep 2025
- IISA International Conference on Statistics, Kochi, India Dec 2024
- External Statistics Conference, Banco de España, Madrid Feb 2024
- SSCA International Conference on Statistics, Banasthali, India Feb 2024
- Reserve Bank Staff College, Chennai, India (virtual) Oct 2023

Contributed Talks/Posters

- 10th Workshop on Biostatistics & Bioinformatics, Atlanta (poster) May 2025
- Symposium on Data Science & Statistics (SDSS), Salt Lake City May 2025
- ISPS International Conference on Statistics, Allahabad, India Feb 2024
- European Meeting of Statisticians (EMS), Poland (poster) Jul 2023
- ENAR Spring Meeting, Nashville, Tennessee Mar 2023
- IISA International Conference on Statistics, Bangalore, India Dec 2022

PROFESSIONAL ACTIVITIES

Journal Peer Review

- Journal of Statistical Theory and Practice
- Journal of Investigative Medicine

Service

- Session Chair: Topics in High-Dimensional and Network Data, IISA Conference 2024
- Contributed to 'State of the Economy' articles, RBI Monthly Bulletin 2022–2023
- Committee Member, Placement Cell, IIT Bombay 2021
- Executive Member, Post Graduate Academic Council, IIT Bombay 2020
- Volunteer for IWM and IISA conference, IIT Bombay 2019
- General Secretary, Hostel 11, IIT Bombay 2019
- Led yoga and wellness workshops, inspiring over 2k to live healthier 2020 - present

Membership

- American Statistical Association (ASA)
- Caucus for Women in Statistics & Data Science (CWS)
- International Indian Statistical Association (IISA)
- Society of Statistics, Computer & Applications (SSCA)

Training & Development

- Statistical Learning with Python, Stanford Online, edX 2025
- Preparing Future Faculty, Auburn University 2025
- Course (Re)Design, Auburn University 2024, 2025
- Compilation of National Accounts, 1-week, NSSTA, Noida, India 2022
- JAIB Certification, Indian Institute of Banking & Finance 2022
- Remittance Statistics, 2-day, IMF's Statistics Department (virtual) 2022
- Trade in Services Statistics, 6-week, UNSD, WTO, and UNCTAD (virtual) 2021
- BOP & IIP Statistics, 3-month, IMF Institute (virtual) 2021
- Power BI Data Analytics, 2-day, RBSC, Chennai, India 2021
- Yoga & Wellness Certifications (500-hr YTT; 1000-hr SKY Coach) 2021–2025
- Mathematics for Data Science, 2-week, IISC Bangalore, India 2019

COMPUTER SKILLS

- Extensive programming experience with R
- Programming experience with Python, Mathematica, and C++,

REFERENCES

Prof. Roberto Molinari

Assistant Professor of Statistics
Auburn University, USA
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Prof. Siuli Mukhopadhyay

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Prof. Sujit Ghosh

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North Carolina State University, USA
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